Functional Requirement Document (FRD)

Change Management Software Module - Pharmaceutical Industry

SmartFoodSafe

Digitalize Food Safety

1. Introduction

This document defines requirements for a **Change Management System** to ensure compliance, traceability, and control over changes in **pharmaceutical processes**.

The system aims to ensure a controlled handling of all changes related to processes, equipment, documents, and systems which ensures Good Manufacturing Practices (GMP).

It enables users to submit, review, approve, and track change requests with complete traceability, audit trails, and risk assessment capabilities. The system is intended to be used by employees, quality assurance teams, and auditors to ensure transparency, accountability, and product quality.

2. Objectives

- Ensure compliance with GMP and regulatory standards
- Provide complete traceability and audit trails
- Enable structured approval workflows
- Improve operational transparency and accountability

3. User Roles

- **Admin** – Manage users and system configurations
- **Requester** – Create and submit change requests
- **Reviewer** – Evaluate change impact
- **Approver (QA)** – Approve or reject changes
- **Auditor** – View logs and reports

4. Functional Requirements

4.1 Authentication

- Secure login with username and password
- Role-based access control
- Session timeout after inactivity

4.2 Change Request Module

- Auto-generate Change ID
- Mandatory fields: title, description, department
- Attach documents required for the change
- Edits allowed before submission only

4.3 Impact Assessment

- **Risk classification:** Low, Medium, High

- Identify impacted systems/departments

4.4 Workflow & Approval

- **Multi-level approval:** Reviewer → QA → Closure
- **Status transitions:** Draft → Submitted → Approved → Implemented → Closed
- Rejection requires comments

4.5 Audit Trail

- Capture all user actions with timestamps

4.6 Notifications

- Email alerts for approvals, rejections, updates
- Dashboard alerts for pending tasks

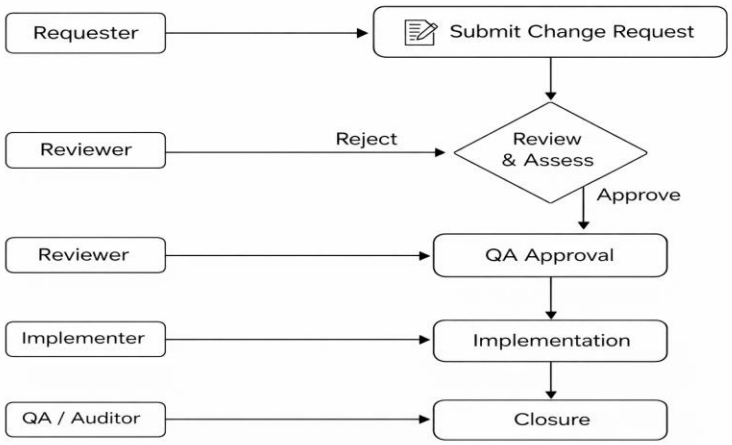
4.7 Compliance Requirements

- Electronic signatures for approvals
- Audit readiness support

5. Acceptance Criteria

- System should not allow submission with missing mandatory fields
- All approvals must be logged with timestamp and user ID
- Audit logs must be non-editable

6. Workflow Diagram



7. Sprint Plan

Sprint 1: Authentication & Dashboard

- Login UI
- Role management
- Dashboard UI

Sprint 2: Change Request Module

- Form UI
- Database schema
- Validation logic

Sprint 3: Workflow & Approval

- Approval engine
- Status tracking
- Notifications

Sprint 4: Audit & Testing

- Audit logs
- Compliance checks
- System testing

8. Mock-up Screens

- **Login Screen** – Username, Password, Login button
- **Dashboard** – Summary cards (Open, Approved, Rejected)
- **Change Request Form** – Input fields and attachment section
- **Approval Screen** – Approve/Reject buttons with comments

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